

Euler Paths

Math 252

An Euler Path Algorithm

Here is python code for an Euler path algorithm.

```
# find an Euler path/circuit or report there is none.
# this version assumes (without checking) that the graph is connected.
def euler_path(graph, verbose = False):
    degrees = graph.degrees()
    odd_vertices = [v for v in degrees.keys() if degrees[v] % 2 == 1]

    if len(odd_vertices) == 2:
        v_init = odd_vertices[0]
    elif len(odd_vertices) == 0:
        v_init = graph.some_vertex()
    else:
        return(None)

    stack = [v_init]
    path = []

    while stack:
        if verbose:
            print "stack =", stack, " path =", path
        u = stack[-1]
        v = graph.adjacent_vertex(u)
        if v:
            stack.append(v)
            graph.delete_edge((u,v))
        else:
            # move vertex from stack to path
            path.append(stack.pop())
    return(path)
```

Examples

```
verbose = False
G1 = Graph([(1,2), (1,3), (1,4), (1,5), (2,3), (2,5),
            (2,6), (3,4), (3,5), (4,5), (4,6)])
print euler_path(G1, verbose = verbose)
```

```
[1, 5, 4, 6, 2, 5, 3, 4, 1, 3, 2, 1]
G2 = Graph([(1,2), (1,3), (1,4), (2,3), (2,5),
            (2,6), (3,4), (3,5), (4,5), (4,6)])
print euler_path(G2, verbose = verbose)
```

```
[5, 4, 6, 2, 5, 3, 4, 1, 3, 2, 1]
G3 = Graph([(1,2), (1,3), (1,4), (2,3), (2,5), (2,6),
            (3,4), (3,5), (4,5), (4,6), (5,6), (3,7)])
print euler_path(G3, verbose = verbose)
```

None

Questions

1. What would the output of `euler_path(G1, verbose = True)` be?
(For this question, you may assume that `adjacent_vertex()` will return the smallest numbered adjacent vertex and `some_vertex()` the smallest numbered vertex in the graph.)
2. Pick a graph representation (edge list, adjacency list, adjacency matrix, incidence matrix) and determine the efficiency of each method of the `Graph` class used in this algorithm.
(For this question, you do not need to return any particular vertex from `adjacent_vertex()` or `some_vertex()`. Which vertex does your algorithm choose?)
3. For your graph representation, determine the efficiency of the algorithm.
Be sure to consider the efficiency of the methods in the `Graph` class.
4. Repeat questions 2 and 3 for other representations.
5. What happens in this algorithm if the graph provided is not connected?
6. Can this algorithm handle multi-edges and self loops?
7. Will the algorithm work for directed graphs?
8. This algorithm assumes the graph is connected. What is the efficiency of checking that a graph is connected?